



## SHANKAR K.G

Backend Developer

### CONTACT

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### EDUCATION

#### B.E. Computer Science

DR. N.G.P. Institute of Technology  
Expected Grad: 2028  
CGPA: 8.56, Current Semester - 5

#### Class XII

A.V.P. Trust National Matric Higher  
Secondary School  
Graduation Year: 2024  
Percentage: 93

#### Class X

A.V.P. Trust National Matric Higher  
Secondary School  
Graduation Year: 2022  
Percentage: 92.2

### TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript
- Frameworks:** Django, Spring Boot
- Database:** PostgreSQL
- Tools:** Git, GitHub, VS Code
- Concepts:** OOP, DBMS, REST APIs

### SOFT SKILLS

- Logical Problem Solving
- Adaptability
- Team Collaboration

### LANGUAGES

- Tamil
- English

## Objective

I am a Computer Science student currently pursuing my undergraduate degree. I am looking for a backend developer role where I can build reliable applications and solve practical problems. I have hands on experience writing clean code, working with databases, and taking ideas from initial concepts to working projects. I am eager to bring my logic skills and positive attitude to a fast paced engineering team.

## Academic Projects

### FinTracker: Personal Finance System

Web Application

*Django, PostgreSQL, JavaScript, HTML5/CSS3*

- Developed a full-stack finance application using Django and PostgreSQL to help users actively track income, expenses, and savings goals.
- Engineered secure user authentication modules and dynamic JavaScript dashboards to visualize monthly spending data.
- Designed a responsive user interface following Django's MVT architecture to ensure a clean, maintainable backend structure.

### CAIP – Karnataka Intelligence Platform

Analytics & Intelligence System

*Python, FastAPI, React, PostgreSQL, Docker*

- Built a web-based dashboard for police agencies to view crime trends and map criminal networks across districts.
- Developed a text processing tool that reads PDF police reports to automatically identify suspects, gang ties, and repeat offenders.
- Designed fast backend APIs for bulk data uploads and set up Docker to run the application components easily.

### ESP32 Wireless Air Mouse

Hardware Integration

*Python, ESP32, Sensor Interfacing, BLE Protocols*

- Engineered a wireless hardware mouse using an ESP32 microcontroller and MPU6050 sensor to translate hand gestures into precise cursor movements.
- Configured Bluetooth Low Energy (BLE) HID protocols to stream sensor telemetry and implemented calibration algorithms to reduce cursor drift.
- Wired hardware buttons for standard mouse clicks and programmed a dedicated toggle switch for motion tracking control.

## Key Achievements

- Completed over 175 coding challenges across LeetCode and GeeksforGeeks to strengthen my understanding of core data structures.
- Earned a verified certification in Django with Python from Pine Sphere after completing practical training on web routing and models.
- Completed the Introduction to Internet of Things academic training through NPTEL.